MATHS MANTHAN
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Subject: Mathematics | **PERIODIC ASSESSMENT TEST – 07 (QUADRATIC EQUATIONS)** | **Time:** 60 Minutes

Class: Class X (CBSE) | **Chapter 4: Quadratic Equations** | **MM: 30**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choice is provided in Question 13(iii).
2. **Section A** comprises 5 MCQs and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 2 Very Short Answer questions (VSAQ) of 2 marks each.
4. **Section C** comprises 2 Short Answer questions (SA) of 3 marks each.
5. **Section D** comprises 2 Long Answer questions (LA) of 5 marks each.
6. **Section E** comprises 1 Applied Case Study with 3 sub-parts of 4 marks total.

SECTION A (Multiple Choice & Assertion-Reason Questions 6 × 1 = 6 Marks)

Q1. The least positive value of k for which the quadratic equation $2x^2 + kx - 4 = 0$ has rational roots is: [1]
(A) $\pm 2\sqrt{2}$ (B) 2 (C) 0 (D) $\sqrt{2}$

Q2. If the sum of the roots of the equation $x^2 - (k+6)x + 2(2k-1) = 0$ is equal to half of their product, then $k =$: [1]
(A) 6 (B) 7 (C) 5 (D) -7

Q3. The discriminant of the quadratic equation $(x+5)^2 = 2(5x-3)$ is: [1]
(A) 124 (B) -124 (C) 31 (D) -31

Q4. If $y = 1$ is a common root of the equations $ay^2 + ay + 3 = 0$ and $y^2 + y + b = 0$, then ab equals: [1]
(A) 3 (B) -3 (C) 4 (D) 3.5

Q5. If the quadratic equation $x^2 + 4x + k = 0$ has real and distinct roots, then: [1]
(A) $k < 4$ (B) $k > 4$ (C) $k \leq 4$ (D) $k \geq 4$

Q6. Assertion-Reason: [1]
Directions: (A) Both A & R true, R explains A; (B) Both true, R does NOT explain A; (C) A true, R false; (D) A false, R true.
Assertion (A): The roots of the quadratic equation $x^2 - 8x + 16 = 0$ are real and equal.
Reason (R): A quadratic equation $ax^2 + bx + c = 0$ has equal roots if its discriminant $D = b^2 - 4ac = 0$.

SECTION B (Very Short Answer Questions 2 × 2 = 4 Marks)

Q7. Find the values of k for which the quadratic equation $kx^2 - 6x - 2 = 0$ has real roots. [2]

Q8. Solve the following quadratic equation for x by factorization: [2]

$$3x^2 - 2\sqrt{6}x + 2 = 0$$

SECTION C (Short Answer Questions $2 \times 3 = 6$ Marks)

Q9. Solve the following equation for x: [3]

$$\frac{x-1}{x-2} + \frac{x-3}{x-4} = 3\frac{1}{3}, \quad x \neq 2, 4$$

Q10. Solve the following equation for x: [3]

$$\frac{x+3}{x+2} = \frac{3x-7}{2x-3}, \quad x \neq -2, \frac{3}{2}$$

SECTION D (Long Answer / HOTS Questions $2 \times 5 = 10$ Marks)

Q11. The sum of the areas of two squares is 468 m^2. If the difference of their perimeters is 24 m, formulate the quadratic equation and determine the lengths of the sides of the two squares. [5]

Q12. A train travels a distance of 480 km at a uniform speed. If the speed had been 8 km/h less, it would have taken 3 hours more to cover the same distance. Find the original uniform speed of the train. [5]

SECTION E (Applied Case Study Problem $1 \times 4 = 4$ Marks)

Q13. Case Study: Architectural Interior Design & Floor Tiling Optimization

A banquet hall has a large rectangular floor that can be completely covered with 200 identical square ceramic tiles of side length x units. During an interior redesign, the project architect discovers that if the side length of each tile is increased by 1 unit, it would require only 128 tiles to pave the entire floor surface completely.

(i) Express the total surface area of the floor in terms of x when tiled using 200 square tiles. [1]

(ii) Set up the quadratic equation in x in standard form equating the two area expressions. [1]

(iii) Find the original side length x of each tile, and calculate the total surface area of the floor. [2]

OR

Determine how many tiles would be needed to pave the floor if tiles of side length $(x+2)$ units were used.

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